

Department of Physics, FNSPE, CTU in Prague
02YMECH - Mechanics
Make-up / Midterm 2

9.1.2026 / 14:00 - 15:30

Q1: Consider a system of two point particles, each having identical masses m , moving on a xy plane with accelerations, $\vec{a}_1 = (0, a)$ and $\vec{a}_2 = (a/2, 0)$ where a , is a positive constant. The particles are initially at rest in positions, $\vec{r}_1(0) = (1, 0)$ and $\vec{r}_2(0) = (0, 2)$.

- (a) Find the total torque about the origin and the total force. — *0.4 points*
- (b) Are total linear momentum and total angular momentum conserved? — *0.2 points*
- (c) Find the total kinetic energy in the LAB and center of mass frames. — *0.4 points*

Q2: The equation of motion of a damped oscillator is given as $x(t) = e^{-\beta t} x_0 \cos(\omega_1 t)$. How many oscillations N does it approximately take for the amplitude of the oscillation to reduce to $1/e$ if the quality factor Q of the damped oscillator is 90? Assume that the oscillator is weakly damped, i.e., $\omega_1 \approx \omega_0$. (Givens: $Q = \omega_0/2\beta$, $T_0 = 2\pi/\omega_0$, $\pi \approx 3$) — *1 point*

Q3: A particle of mass μ moves under a central force $f(r)$ with angular momentum l . Its orbit is given by $r(\theta) = \frac{1}{a\theta}$ where $a > 0$. The orbit of a particle under a central force satisfies

$$\frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) + \frac{1}{r} = -\frac{\mu}{l^2} r^2 f(r).$$

- (a) Find the central force $f(r)$ in terms of μ , l , and r . — *0.6 point*
- (b) Find the potential $U(r)$ associated with this force. — *0.4 point*

Q4: A wagon of mass m moves without friction along a straight, rigid rail fixed to a horizontal platform rotating with constant angular velocity Ω . The rail is oriented radially outward from the axis of rotation and rotates together with the platform. Initially, the wagon is held at a distance r_0 from the axis of rotation and is at rest relative to the rail. It is then released and allowed to move freely along the rail.

- (a) In the reference frame rotating with the platform, write down the equation of motion for the wagon, including all inertial forces. — *0.5 point*
- (b) Solve the equation of motion to obtain $r(t)$. — *0.2 point*
- (c) Determine the time it takes for the wagon to reach the end of the rail located at radius $R > r_0$. — *0.2 point*
- (d) Explain physically why the wagon accelerates outward even though its initial velocity in the rotating frame is zero. — *0.1 point*

Givens:

- Coriolis force: $-2m(\vec{\Omega} \times \vec{v}')$
- Centrifugal force: $-m\vec{\Omega} \times (\vec{\Omega} \times \vec{r}')$
- $\cosh(x) = \frac{e^x + e^{-x}}{2}$
- Solution hypothesis for the equation $\ddot{x} \pm \omega^2 x = 0$ is $x(t) = e^{\lambda t}$.

